Code No: 131AE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year I Semester Examinations, March/April - 2023

ENGINEERING MECHANICS

(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, AE, MIE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) State and explain Varignon's Theorem. [2]
b) State the general algebraic conditions of equilibrium for concurrent-coplanar forces and concurrent - non-coplanar forces. [3]
c) Define the term angle of repose. [2]
d) A body of 500-N weight is resting on a rough horizontal surface. Find the magnitude of the pulling force acting on the body at an angle of 30° with the horizontal, which will just make the body move. Take $\mu = 0.3$. [3]
e) What is the difference between the centre of gravity, centre of mass and centroid? [2]
f) State and prove the parallel axis theorem. [3]
g) Explain the procedure to find polar moment of inertia of masses. [2]
h) What are the mass moments of inertia of a uniform slender rod of length l when it is supported at its one of the end? [3]
i) State principle of conservation of momentum and list its applications. [2]
j) A 5-kg block drops from 2 m upon a spring whose modulus is 10,000 N/m. What will be the speed of the block when the spring is deformed 100 mm? [3]

PART - B

(50 Marks)

2.a) What is a free body diagram? Explain with a suitable example.
b) Two systems of forces are shown in figure 1. Are they equivalent? [4+6]

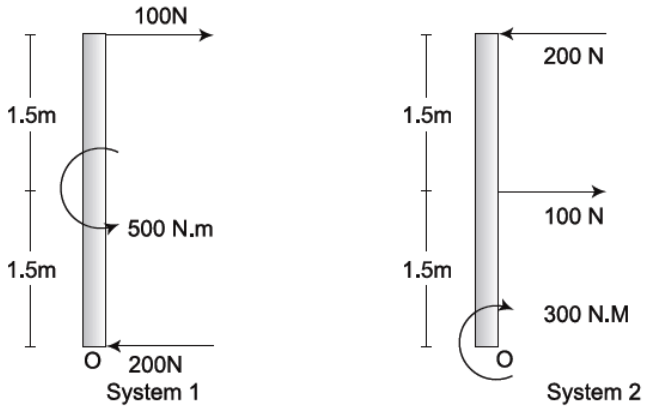


Figure 1

OR

3. The beam EF in figure 2 weighs 120 N/m and carries the 600-N weight at its end. It is supported by a ball-and-socket joint at E and the cables AB and CD. Determine the tensions in AB and CD. Find the joint reactions at E. All dimensions are in cm. [10]

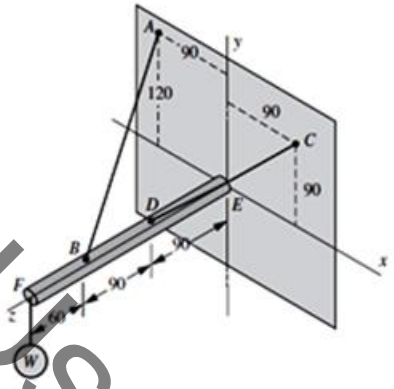


Figure 2

4. Determine the minimum angle at which a uniform ladder can be placed against a wall without slipping under its own weight. The coefficient of friction for all surfaces is 0.2. What will be this angle if both the walls are taken as smooth surfaces? [10]

OR

5. A 15^0 wedge has to be driven downwards for moving the block B of 750 N weight as shown in figure 3. If the angle of friction for all surfaces is 10°, find the force P required to move the wedge. What will be the value of P if friction on horizontal floor and vertical wall neglected? The coefficient of friction between the wedge and block remain same. [10]

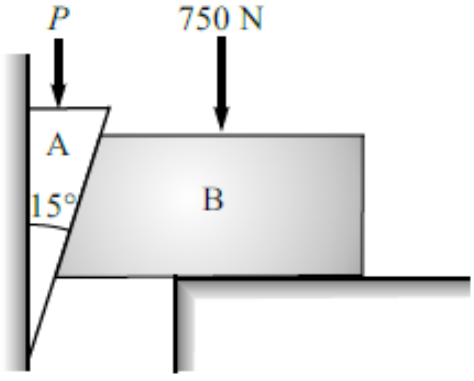


Figure 3

6. A right circular cone of 300 mm height and a base of 300 mm diameter rest on the top of a right circular cylinder of the same radius and 300 mm height. Locate the centroid of the combined volume if the densities of the two are the same. What happens to the centroid if the material density of the cylinder is twice of that of the cone? [10]

OR

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7. Calculate the moment of inertia of the composite area shown in figure 4 about the two centroidal axes and its polar moment of inertia. [10]

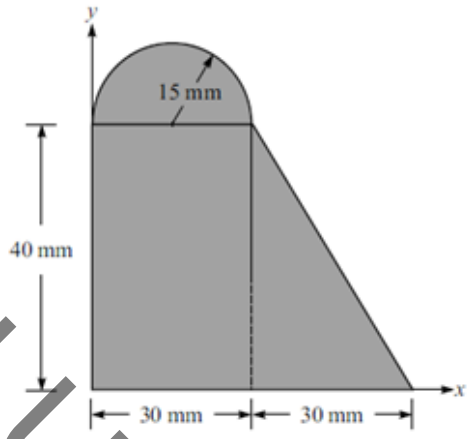


Figure 4

8. Using the integration approach, derive the mass moment of inertia of a rectangular plate about X-X axis passing through the CG of the plate. Use this expression to find the mass moment of inertia about the base of the plate. [10]

OR

9. Compute the moment of inertia of the cast-iron flywheel shown in figure 5. The density of cast iron is 7200 kg/m^3. All dimensions are mm. [10]

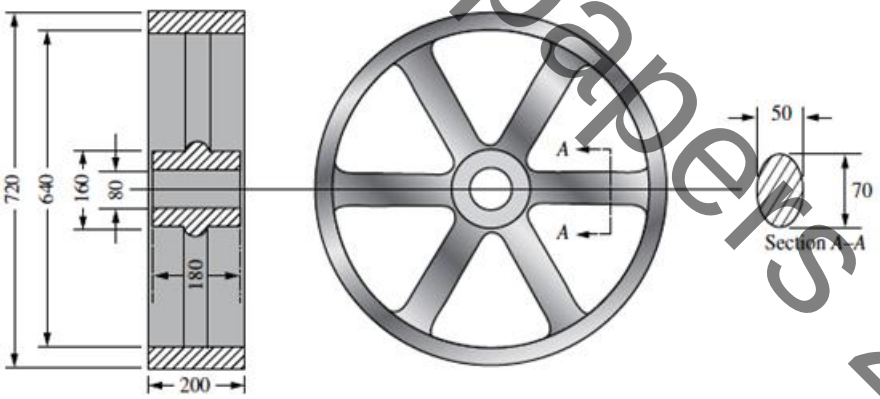


Figure 5

10.a) A link AB is moving in a vertical plane. At a certain instant, when the link is inclined at 60° to the horizontal, the point A is moving horizontally at 5 m/s, while B is moving vertically upwards. Find the velocity of B.

b) A cylinder of radius 1 m rolls without slipping along a horizontal plane AB. M is the contact point of the plane and the cylinder. N is the point on the other end of diameter passing through M. Find the velocity of point M and N. Its centre has uniform velocity of 20 m/s. [5+5]

OR

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11.a) A particle of mass 1 kg moves in a straight line under the influence of a force which increases linearly with time at the rate of 60 N per sec. At time t = 0, the initial force may be taken as 50 N. Determine the acceleration and velocity of the particle 4 sec after it started from rest at the origin.

b) A bullet of mass 25 gm, moving horizontal with a velocity of 600 m/s strikes a wooden block of mass 5 kg resting on a rough horizontal surface. The bullet after striking the block remains buried in the block and both travels together. Find the common velocity of bullet and block after impact. [6+4]

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